Mobile Devices : Project Proposal - Rough Draft

What needs to be included in the proposal:

- Overview of project (1)
- List of group members + responsibilities (1)
- List of features (functional requirements) & usage (2)
- Code design (e.g., UML) (2)
- Mockup of UI (2)
- Writing quality (1)
- Scope reasonable for team size (1)

List of Group Members & Responsibilities:

Aakanksha Parekh

- ➔ Created project proposal - rough draft
- ➔ Overview Section
- ➔ Feature List

Sanhith Amarathunge

- ➔ Feature List

Paul Joseph

Sunny Patel

- ➔ Code Design - Technical Implementations for Flutter/Dart
- ➔ UML - cohesive of Technical Draft

Overview:

MindEase is a mobile app created to assist student mental health and overall wellness in a progressively challenging academic setting. The app provides a secure, intuitive environment for students to track their emotions, journal their thoughts, and participate in structured wellness practices like breathing exercises, brief meditations, and mindfulness tasks.

In addition to fundamental tracking, MindEase offers tailored insights via data visualization and trend analysis, assisting students in comprehending their emotional patterns over time more effectively. Regular prompts and inspirational alerts foster persistence in self-care routines, while resource connections direct users to professional mental health services and hotlines for extra assistance when necessary.

The application prioritizes simplicity, accessibility, and privacy, allowing students to utilize it instantly without requiring specialized equipment or intricate configurations. All primary functions operate offline with local storage, while extra benefits come from real-time wellness tips and quotes retrieved from external APIs.

To a user, MindEase seems more like a personal companion than just an app—constantly available to offer support, monitor progress, and encourage healthier coping methods. Its sleek layout, user-friendly navigation, and insights based on data render it both useful and enjoyable for everyday use.

In the end, MindEase sees itself as more than merely a tool for productivity; it serves as a wellness platform for students that fosters resilience, self-awareness, and mental equilibrium in today's rapid academic environment

List of Features & Usage:

Storage:
- ➔ Storing Student/User well-being notes, conditions and concerns
- ➔ Storing their journals
- ➔ Data Export (JSON/ZIP formats)/Delete Functionality

Multiple Screens:
- ➔ Dashboard
- ➔ Account settings
- ➔ Activities
- ➔ Tracker
- ➔ Crisis & Resources

Geo Location, Geocoding, maps:
- ➔ Help near me
    - ◆ User enters postal code (Optional, the app will **not** continuously track the user's location)
    - ◆ Direct Call/Text buttons for hotlines
    - ◆ Display map of closest hospitals, and institutes
    - ◆ When user selects one location, it will show human address and geocode

Camera:
- ➔ Profile Picture
- ➔ What I did Today/Journaling
    - ◆ User can upload photo of what they want to share in their journal
    - ◆ Guided Prompts
    - ◆ Voice Notes

HTTP Request:
- ➔ Fetch Quotes & Wellness tips from external API

Internalization:
- ➔ Can add English and French

Data Table and charts:
- ➔ Weekly Reports and Stats of user
- ➔ Analyze user trends
    - ◆ App will potentially provide actionable insights to user habits

➔ Zero-Pressure Streak system (user can pause the streak whenever they want)

Notification:

➔ “How are you feeling today?”
➔ Journal your day!
➔ Reminder to eat
➔ Reminder to take a walk

Dialogs & Pickers

➔ Mood picker (emojis/scale) for daily check-ins.
➔ Date/time pickers for setting reminders.

Snack Bars

➔ Quick feedback after saving or deleting an entry.
➔ Undo option for accidental deletions.
➔ Alerts for issues (e.g., no internet connection).

Code Design:

Code Design (Sunny Patel)

Platform & Framework

- Language & Framework: Flutter with Dart.
- UI Style: Material 3 as the base design system. Cupertino components may be used on iOS where it improves user experience.
- Theming: Light and dark modes supported. Semantic colors mapped to mood states.

Architecture

- Layered Approach:
  • UI Layer: Screens and widgets.
  • Repository Layer: Handles data access and abstracts Firebase calls.
  • Service Layer: Interfaces for notifications, location, and media.
- Navigation: Named routes for each screen (Dashboard, Daily Check-in, Journal, Activities, Crisis & Resources, Settings).

Data & Storage

- Backend: Firebase (Firestore + Storage).
- Collections:
  • Users and UserSettings
  • MoodCheckins
  • JournalEntries (with optional Assets)
  • Activities and ActivitySessions
  • Reminders

  • Resources and ResourceLocations
  • Tags (optional, for organizing journal entries)
- Media: Firebase Storage for photos/audio; Firestore stores only metadata + URLs.
- Export/Delete: JSON export of user-owned data; clear “delete my data” flow.

Domain Model (Aligned to UML)
- User → UserSettings: 1:1 relationship.
- User → MoodCheckin / JournalEntry / ActivitySession / Reminder / StreakPause: 1:N relationships.
- JournalEntry → JournalAsset: 1:N for photos/audio.
- JournalEntry ↔ Tag: M:N using a reference array.
- Activity → ActivitySession: 1:N.
- Resource → ResourceLocation: 1:N.

Services
- Notification Service: Local notifications (once, daily, weekly).
- Location Service: Postal code entry + optional geocoding. No background tracking.
- Media Service: Camera/gallery + audio recording.
- Export Service: Builds JSON export.
- Intl Service: English + French.

State Management
- Riverpod or Provider for lightweight state handling.
- Screens listen to repository streams for updates.
- Fallbacks: Denied permissions show friendly fallbacks.

Validation & Constraints
- Mood score must be 1–5.
- Journal requires text; attachments optional.
- Activity session must end after start.
- Location optional, must be within valid lat/long if present.
- Reminders limited to simple recurrence (once/daily/weekly).

Charts & Insights
- Weekly charts for mood and activity adherence.
- Data aggregated locally from cache.
- Streak pause windows supported.

Internationalization & Accessibility
- Starts in English;
- Screen reader labels and large tap targets.
- High-contrast themes.

Error Handling

- Inline, human-readable messages.
- Network-optional features fail gracefully.
- Undo snackbars for destructive actions.

Testing & Scope
- Unit Tests: Repositories and validators.
- Widget Tests: Daily Check-in and Journal entry flows.
- Golden Tests: Dashboard and charts.

Build Plan (Milestones)
1. Project skeleton, navigation, theming.
2. Mood Check-in flow.
3. Journal with photo/audio.
4. Activities catalog and sessions.
5. Local reminders.
6. Resources screen.
7. Export/Delete.
8. Optional features (quotes API, postal code search, streak visualization).

Code Design EXTENDED UML (Sunny):

Below is a rough UML model of what we plan to accomplish with this application model (correlant to the Code Design & Overview)

